

Robbe De Win

3000 Leuven, Belgium | +32 472 81 31 97 | robbedewin@hotmail.com | linkedin.com/in/robbedewin | github.com/robbedewin

Profile

Bioinformatician with two completed master's degrees, most recently building automated genome-scale metabolic modelling pipelines across roughly 1,500 bacterial genomes. Comfortable owning a computational pipeline end to end — reconstruction, quality control, validation against experimental data, and the reproducibility discipline that makes results trustworthy. Python and R daily, Linux and Git as everyday tools.

Education

Master of Bioinformatics

2024 – 2026

KU Leuven

- Result: Cum Laude
- Thesis: *Pan-Metabolic Modelling Software for Therapeutics*
- Relevant coursework: Machine Learning, Artificial Neural Networks, Bayesian Modelling, Comparative & Regulatory Genomics, Evolutionary and Quantitative Genetics

Master of Biomedical Sciences

2022 – 2024

KU Leuven

- Result: Cum Laude
- Thesis: *Genetic Changes and Allele-Specific Expression in Induction Failure T-Cell Acute Lymphoblastic Leukemia*

Bachelor of Biomedical Sciences

2019 – 2022

KU Leuven

- Result: Magna Cum Laude

Research

MSc Thesis — Pan-Metabolic Modelling Software for Therapeutics

2025 – 2026

KU Leuven

- Built PaRMMoSahn, an open-source Python package that reconstructs one pan-metabolic model per species (PPanGGOLiN pangenome searched with gapseq) and derives every strain model from it by DIAMOND protein-homology projection — minutes per genome instead of hours.
- Validated on 1,010 *Clostridia* and 481 *Bacillus subtilis* genomes plus seven in-house lab strains, scored against Biolog substrate-utilisation data as Matthews correlation coefficient.
- Matched or exceeded per-genome reconstruction by gapseq and CarveMe in every cohort, with the highest pooled MCC on eight held-out *B. subtilis* strains, at roughly 12× the throughput on the largest cohort.
- Model QC with memote; downstream flux balance analysis, gene essentiality and auxotrophy prediction.
- Code: github.com/robbedewin/PaRMMoSahn

MSc Thesis — Genetic Changes and Allele-Specific Expression in Induction Failure T-ALL

2023 – 2024

KU Leuven

- Built Snakemake WGS and RNA-seq pipelines on the telomere-to-telomere T2T-CHM13 reference to profile induction-failure T-ALL patient samples: somatic and structural variant calling (Mutect2, GRIDSS) and copy-number analysis (ASCAT).

- Implemented a beta-binomial test for allele-specific expression combining DNA-derived allele counts with RNA read counts. Recovered known *NOTCH1* and *CDKN2A/B* lesions and identified *FAT1* as showing strong allele-specific expression.

Technical skills

Languages: Python, R, Bash; LaTeX

Bioinformatics: Genome-scale metabolic modelling (gapseq, CarveMe, memote), flux balance analysis, pangenome analysis (PPanGGOLiN, DIAMOND), gene essentiality and auxotrophy prediction; WGS/RNA-seq variant calling (Mutect2, GRIDSS, ASCAT), allele-specific expression

Workflows: Snakemake, conda, reproducible staged pipelines

Data & ML: Machine learning and neural networks, Bayesian modelling, high-dimensional data analysis, statistical validation

Engineering: Git/GitHub, Linux, conda environment management, pytest, Quarto, reproducible pipeline design, long-running batch workloads

Software projects

financial_report

github.com/robbedewin/financial_report

Quarto-based financial planning report generator: templated reports, net-worth and snapshot modules, an audit tool with alerting, and a pytest suite.

market_data

github.com/robbedewin/market_data

ISIN-keyed instrument registry and price-series store with importers and downstream projections feeding financial_report.

Work experience

Student summer job

2017 – 2025

Belcar (automotive)

- Nine consecutive summers with the same employer: vehicle handling and on-site movement, plus administrative support.

Delivery Driver (student)

2020 – 2022

B.B. Trans

Languages

Dutch: Native

English: Full professional proficiency

French: Basic